```
/Users/mehakagrawal/PycharmProjects/Final_Dissertation/.venv/bin/python /Users/mehakagrawal/Desktop/Final_Dissertation/Codes/3.go_enrichment_analysis/go_enrichment_analysis.py
Loading data...
Significance column in use: qvalue (cutoff 0.05, |log2FC|>1.0)
Q1 DEGs: 5541 (1289 up, 4252 down)
Q2 DEGs: 515 (368 up, 147 down)


STEP 1: GO / Pathway Enrichment (gProfiler)

  Running enrichment for Q1 Upregulated in MUT (1289 genes) (1289 genes)...
  Found 154 significant terms
    [KEGG] Cytoskeleton in muscle cells (n=61, p=8.12e-23)
    [GO:CC] membrane (n=259, p=2.86e-09)
    [GO:CC] extracellular region (n=66, p=1.65e-08)
    [GO:CC] cell periphery (n=91, p=2.40e-08)
    [GO:BP] external encapsulating structure organization (n=14, p=5.84e-07)
    [GO:BP] extracellular matrix organization (n=14, p=5.84e-07)
    [GO:BP] extracellular structure organization (n=14, p=5.84e-07)
    [GO:MF] protein binding (n=322, p=1.30e-06)
    [GO:CC] sarcoplasm (n=7, p=1.53e-06)
    [GO:CC] extracellular space (n=30, p=8.91e-06)
    [GO:CC] plasma membrane (n=77, p=2.12e-05)
    [GO:CC] sarcoplasmic reticulum (n=5, p=4.01e-05)
    [GO:CC] actin cytoskeleton (n=22, p=4.01e-05)
    [GO:CC] contractile muscle fiber (n=11, p=5.39e-05)
    [GO:CC] myofibril (n=11, p=5.39e-05)

  Running enrichment for Q1 Downregulated in MUT (4252 genes) (4252 genes)...
  Found 346 significant terms
```

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    [GO:BP] microtubule-based process (n=126, p=5.43e-33)
    [GO:BP] cell cycle process (n=97, p=9.30e-33)
    [GO:BP] DNA metabolic process (n=140, p=7.70e-31)
    [GO:BP] cell cycle (n=114, p=4.74e-30)
    [GO:MF] adenyl ribonucleotide binding (n=405, p=4.53e-29)
    [GO:MF] ATP binding (n=404, p=4.53e-29)
    [GO:MF] adenyl nucleotide binding (n=410, p=7.92e-27)
    [GO:CC] microtubule cytoskeleton (n=81, p=6.73e-24)
    [GO:BP] microtubule-based movement (n=68, p=7.12e-23)
    [GO:MF] binding (n=1862, p=2.51e-21)
    [GO:BP] DNA damage response (n=96, p=2.79e-21)
    [GO:CC] chromosome (n=111, p=3.25e-21)
    [GO:BP] DNA repair (n=88, p=1.31e-20)
    [GO:BP] DNA replication (n=47, p=2.43e-19)
    [GO:CC] condensed chromosome (n=32, p=3.21e-19)

  Running enrichment for Q2 Upregulated in Old (368 genes) (368 genes)...
  Found 108 significant terms
    [GO:CC] extracellular region (n=61, p=3.74e-36)
    [GO:MF] endopeptidase inhibitor activity (n=32, p=1.37e-29)
    [GO:MF] endopeptidase regulator activity (n=32, p=1.37e-29)
    [GO:MF] peptidase inhibitor activity (n=32, p=9.87e-29)
    [GO:MF] peptidase regulator activity (n=32, p=2.83e-28)
    [GO:MF] enzyme inhibitor activity (n=33, p=2.09e-26)
    [GO:CC] extracellular space (n=35, p=5.29e-26)
    [GO:MF] molecular function inhibitor activity (n=33, p=8.05e-26)
    [GO:MF] enzyme regulator activity (n=36, p=8.21e-12)
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    [GO:MF] serine-type endopeptidase inhibitor activity (n=15, p=9.00e-12)
    [GO:BP] lipid transport (n=18, p=1.88e-11)
    [GO:BP] lipid localization (n=18, p=1.88e-11)
    [GO:MF] molecular function regulator activity (n=46, p=6.77e-11)
    [GO:BP] lipoprotein metabolic process (n=8, p=1.05e-07)
    [GO:BP] coagulation (n=9, p=6.80e-06)

  Running enrichment for Q2 Downregulated in Old (147 genes) (147 genes)...
  Found 6 significant terms
    [KEGG] Arachidonic acid metabolism (n=3, p=2.76e-02)
    [KEGG] VEGF signaling pathway (n=3, p=2.77e-02)
    [KEGG] alpha-Linolenic acid metabolism (n=2, p=2.88e-02)
    [KEGG] Arginine biosynthesis (n=2, p=2.88e-02)
    [REAC] Signaling by Activin (n=2, p=2.92e-02)
    [KEGG] Linoleic acid metabolism (n=2, p=3.41e-02)

  Running enrichment for Q3 Overlap (180 genes) (180 genes)...
  Found 73 significant terms
    [GO:CC] extracellular region (n=29, p=5.36e-15)
    [GO:CC] extracellular space (n=20, p=5.36e-15)
    [GO:MF] endopeptidase inhibitor activity (n=14, p=2.44e-11)
    [GO:MF] endopeptidase regulator activity (n=14, p=2.44e-11)
    [GO:MF] peptidase inhibitor activity (n=14, p=4.26e-11)
    [GO:MF] peptidase regulator activity (n=14, p=5.50e-11)
    [GO:MF] enzyme inhibitor activity (n=15, p=6.16e-11)
    [GO:MF] molecular function inhibitor activity (n=15, p=9.88e-11)
    [GO:BP] lipid transport (n=11, p=5.58e-07)
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    [GO:BP] lipid localization (n=11, p=5.58e-07)
    [GO:CC] high-density lipoprotein particle (n=3, p=2.98e-05)
    [GO:MF] oxidoreductase activity (n=18, p=3.22e-05)
    [GO:CC] protein-lipid complex (n=3, p=3.71e-05)
    [GO:CC] plasma lipoprotein particle (n=3, p=3.71e-05)
    [GO:CC] lipoprotein particle (n=3, p=3.71e-05)
  Saved enrichment_q1_up.csv
  Saved enrichment_q1_down.csv
  Saved enrichment_q2_up.csv
  Saved enrichment_q2_down.csv
  Saved enrichment_overlap.csv
  Saved enrichment_q1_up.png
  Saved enrichment_q1_down.png
  Saved enrichment_q2_up.png
  Saved enrichment_q2_down.png
  Saved enrichment_overlap.png


STEP 2: Transcription Factor Analysis

Q1 DEGs that are transcription factors: 380 out of 5541
Q2 DEGs that are transcription factors: 31 out of 515

Q1 - Top TF families affected by genotype:
  zf-C2H2                         total=121  (↑25 ↓96)
  Homeobox                        total=58  (↑9 ↓49)
  ZBTB                            total=29  (↑13 ↓16)
  bHLH                            total=18  (↑6 ↓12)
  Fork head                       total=14  (↑1 ↓13)
  TF_bZIP                         total=13  (↑7 ↓6
```

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)
  HMG                              total=12  (↑2 ↓10)
  MYB                              total=11  (↑1 ↓10)
  T-box                            total=9  (↑2 ↓7)
  IRF                              total=7  (↑2 ↓5)
  THAP                             total=7  (↑0 ↓7)
  ETS                              total=6  (↑2 ↓4)
  THR-like                         total=5  (↑4 ↓1)
  zf-C2HC                          total=4  (↑0 ↓4)
  E2F                              total=4  (↑0 ↓4)

Q1 - Top 20 DE transcription factors (by significance):
  rcor2                   family=MYB                log2FC=-6.06  qvalue=5.44e-20
  foxm1                   family=Fork head          log2FC=-5.35  qvalue=2.15e-19
  mybl2b                  family=MYB                log2FC=-2.71  qvalue=8.18e-14
  rfx3                    family=RFX                log2FC=-6.06  qvalue=2.21e-13
  sox19b                  family=HMG                log2FC=-4.89  qvalue=4.27e-13
  e2f5                    family=E2F                log2FC=-5.09  qvalue=5.05e-13
  terf1                   family=MYB                log2FC=-3.07  qvalue=1.03e-11
  lhx1a                   family=Homeobox           log2FC=-4.00  qvalue=1.39e-11
  pias4b                  family=zf-MIZ             log2FC=-6.24  qvalue=3.25e-11
  e2f8                    family=E2F                log2FC=-4.28  qvalue=3.87e-10
  hsf5                    family=HSF                log2FC=-11.73  qvalue=4.08e-10
  sall4                   family=zf-C2H2            log2FC=-2.64  qvalue=8.60e-10
  klhl10a                 family=ZBTB               log2FC=-6.14  qvalue=8.70e-10
```

```
  pou2f2a              family=Pou            log2FC=-5.58  qvalue=2.10e-09
  wdhd1                family=HMG            log2FC=-2.54  qvalue=2.28e-09
  klhl26               family=ZBTB          log2FC=-2.19  qvalue=1.01e-08
  zbtb14               family=ZBTB          log2FC=-1.36  qvalue=4.12e-08
  znf217               family=zf-C2H2    log2FC=-1.87  qvalue=6.30e-08
  rfx1a                family=RFX            log2FC=-4.69  qvalue=7.47e-08
  sox8b                family=HMG            log2FC=-3.78  qvalue=7.64e-08

Q2 - Top TF families affected by age:
  zf-C2H2                              total=12  (↑4 ↓8)
  Fork head                            total=3  (↑2 ↓1)
  Homeobox                             total=2  (↑1 ↓1)
  zf-GATA                              total=2  (↑2 ↓0)
  MH1                                  total=2  (↑0 ↓2)
  bHLH                                 total=2  (↑2 ↓0)
  HMG                                  total=1  (↑0 ↓1)
  SF-like                              total=1  (↑1 ↓0)
  STAT                                 total=1  (↑0 ↓1)
  MBD                                  total=1  (↑0 ↓1)
  zf-LITAF-like                        total=1  (↑1 ↓0)
  TF_bZIP                              total=1  (↑1 ↓0)
  THR-like                             total=1  (↑0 ↓1)
  RFX                                  total=1  (↑0 ↓1)

Q2 - Top 20 DE transcription factors (by significance):
  znf1179              family=zf-C2H2    log2FC=-5.88  qvalue=1.07e-17
  CABZ01002768.1       family=zf-C2H2    log2FC=-5.13  qvalue=4.32e-06
  nr5a5                family=SF-like    log2FC=+9.02  qvalue=1.89e-05
  LO018181.1           family=zf-C2H2
```

```
                    log2FC=-4.78  qvalue=7.80e-05
  si_ch211-212k5.1      family=zf-C2H2
                    log2FC=+3.62  qvalue=1.61e-04
  vdrb                  family=THR-like
                   log2FC=-2.42  qvalue=3.60e-04
  gata6                 family=zf-GATA
                    log2FC=+4.03  qvalue=3.38e-03
  smad3a                family=MH1
                        log2FC=-3.54  qvalue=3.62e-03
  si_ch211-202h22.9     family=zf-LITAF-like
              log2FC=+4.43  qvalue=4.79e-03
  her4.2                family=bHLH
                       log2FC=+4.98  qvalue=6.63e-03
  znf1048               family=zf-C2H2
                    log2FC=+4.97  qvalue=8.41e-03
  nfil3-3               family=TF_bZIP
                    log2FC=+6.34  qvalue=8.80e-03
  sox7                  family=HMG
                        log2FC=-1.54  qvalue=1.03e-02
  smad6a                family=MH1
                        log2FC=-2.00  qvalue=1.04e-02
  znf977                family=zf-C2H2
                    log2FC=+5.57  qvalue=1.05e-02
  BX005417.2            family=zf-C2H2
                    log2FC=-2.10  qvalue=2.09e-02
  stat6                 family=STAT
                       log2FC=-1.68  qvalue=2.33e-02
  znf1082               family=zf-C2H2
                    log2FC=-4.58  qvalue=2.33e-02
  si_dkey-262g12.7      family=zf-C2H2
                    log2FC=-4.33  qvalue=2.43e-02
  foxa3                 family=Fork head
                  log2FC=+4.20  qvalue=2.47e-02

Transcription factors significant in BOTH Q1 and Q2: 8
  nr5a5                 family=SF-like
                    Q1=-3.58  Q2=+9.02  [OPPOSITE]
  gata4                 family=zf-GATA
                    Q1=-2.18  Q2=+3.67  [OPPOSITE]
  gata6                 family=zf-GATA
```

```
                Q1=-1.51  Q2=+4.03  [OPPOSITE]
  rfx4                  family=RFX                    Q1=-3.36  Q2=-3.18  [SAME]
  si_ch211-202h22.9     family=zf-LITAF-like          Q1=-2.08  Q2=+4.43  [OPPOSITE]
  foxa2                 family=Fork head              Q1=-3.80  Q2=+5.40  [OPPOSITE]
  foxa3                 family=Fork head              Q1=-2.21  Q2=+4.20  [OPPOSITE]
  LO018181.1            family=zf-C2H2                Q1=-4.08  Q2=-4.78  [SAME]

  Saved tf_families.png
  Saved q1_transcription_factors.csv
  Saved q2_transcription_factors.csv


STEP 3: Gene Correlation Analysis
Computing pairwise correlations for top 50 genes...
  Saved correlation_top50.png
  Saved correlation_matrix_top50.csv

  Strongly correlated gene pairs (|r| > 0.85): 775

  Top 20 correlated pairs:
    g2e3             ↔ esco2             r=+0.986  (co-expressed)
    nuf2             ↔ si_dkeyp-117h8.4  r=+0.985  (co-expressed)
    kif11            ↔ kifc1             r=+0.977  (co-expressed)
    si_dkey-25o16.4  ↔ kifc1             r=+0.976  (co-expressed)
    CABZ01058261.1   ↔ kif11             r=+0.976  (co-expressed)
    rad54l           ↔ CABZ01058261.1    r=+0.974  (co-expressed)
    rad54l           ↔ kif11             r=+0.974  (co-expressed)
    rad54l           ↔ nusap1            r=+0.974  (
```

```
co-expressed)
    cdk2            ↔ spdya            r=+0.971  (co-expressed)
    kifc1           ↔ nusap1           r=+0.971  (co-expressed)
    parpbp          ↔ ift27            r=+0.971  (co-expressed)
    si_dkey-25o16.4 ↔ kif11            r=+0.968  (co-expressed)
    rad54l          ↔ lin9             r=+0.967  (co-expressed)
    rtkn2a          ↔ g2e3             r=+0.967  (co-expressed)
    cep70           ↔ lin9             r=+0.967  (co-expressed)
    rad54l          ↔ kifc1            r=+0.965  (co-expressed)
    nuf2            ↔ si_ch211-126c2.4  r=+0.965 (co-expressed)
    ska1            ↔ nuf2             r=+0.964  (co-expressed)
    rtkn2a          ↔ cdk2             r=+0.964  (co-expressed)
    si_dkeyp-117h8.4 ↔ si_ch211-126c2.4  r=+0.963 (co-expressed)
  Saved correlated_gene_pairs.csv


STEP 4: Enhanced Overlap Analysis (Q3)

180 genes are DE in both Q1 (genotype) and Q2 (age)
  Same direction: 16
  Opposite direction: 164
  Saved overlap_fc_scatter.png
  Saved q3_overlap_detailed.csv


STEP 5: Filtering Exploration

Q1 (Genotype) - DEG counts at different thresholds
```

```
:
  Threshold                        DEGs     Up   Down
  -------------------------  ------  -----  -----
  |log2FC|>0.5, qvalue<0.05    7716   2874   4842
  |log2FC|>1.0, qvalue<0.05    5541   1289   4252
  |log2FC|>1.5, qvalue<0.05    4069    474   3595
  |log2FC|>2.0, qvalue<0.05    3238    229   3009
  |log2FC|>2.5, qvalue<0.05    2619    135   2484
  |log2FC|>3.0, qvalue<0.05    2131     98   2033

Q2 (Age) - DEG counts at different thresholds:
  Threshold                        DEGs     Up   Down
  -------------------------  ------  -----  -----
  |log2FC|>0.5, qvalue<0.05     519    369    150
  |log2FC|>1.0, qvalue<0.05     515    368    147
  |log2FC|>1.5, qvalue<0.05     496    361    135
  |log2FC|>2.0, qvalue<0.05     470    345    125
  |log2FC|>2.5, qvalue<0.05     443    334    109
  |log2FC|>3.0, qvalue<0.05     406    312     94

Recommended strict filter (qvalue<0.01, |log2FC|>2):
  Q1: 2386 genes
  Q2: 301 genes
  Overlap: 53 genes


COMPLETE SUMMARY

DE Analysis:
  Q1 (Genotype: MUT vs WT at 18mo): 5541 DEGs
    -> Dominated by downregulation in mutants (4252 down vs 1289 up)
    -> Top gene by significance: kif20a
  Q2 (Age: 37mo vs 11mo in WT): 515 DEGs
    -> 368 up vs 147 down in old fish
  Q3 (Overlap): 180 genes shared
    -> 164 OPPOSITE direction (91%)
    -> 16 same direction (9%)

Enrichment (gProfiler):
```

```
  -> See enrichment CSVs and plots for pathway details

Transcription Factors:
  -> Q1: 380 TFs affected by genotype
  -> Q2: 31 TFs affected by age

Correlation:
  -> 775 strongly correlated gene pairs (|r|>0.85)

Output files:
  enrichment_q1_up.csv/png, enrichment_q1_down.csv/png
  enrichment_q2_up.csv/png, enrichment_q2_down.csv/png
  enrichment_overlap.csv/png
  tf_families.png, q1/q2_transcription_factors.csv
  correlation_top50.png, correlation_matrix_top50.csv
  correlated_gene_pairs.csv
  overlap_fc_scatter.png, q3_overlap_detailed.csv

Go enrichment/KEGG analysis complete!

Process finished with exit code 0
```